

Superconductivity and Disorder

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Preface

This dissertation is the original work of Karan. All external research, software, and code used have been properly cited. The author independently developed the dataset using stochastic mathematical equations, and the software packages applied to analyze the dataset have been appropriately cited.

Abstract

Superconductivity, characterized by zero electrical resistance, is significantly influenced by disorder in the form of material impurities. This study uses a stochastic equation derived from BCS theory to generate the dataset that will help investigate the relationship between disorder in the coupling constant (λ_0) and the critical temperature (t_c). The disorder in the Coupling constant is drawn out from normal distribution with a variance of $\sigma^2 = 0.1$. Our general analysis reveals that disorder increases the critical temperature until it reaches a saturation point at $\lambda_0 = 0.3657$, beyond which a significant decline in t_c is observed. To validate and deepen our analysis, four machine learning models—Random Forest, Gaussian Processes Regression (GPR), Monte Carlo with Linear Regression, and Variational Autoencoder (VAE)—were employed. The results from these models indicate that higher randomness at lower coupling constants ($\lambda_0 \in [0, 0.35]$) positively impacts the critical temperature. These findings provide valuable insights, suggesting that controlled disorder in superconductors can enhance critical temperatures.

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Chapter 1

Introduction

1.1 Superconductivity- Brief Overview

Superconductivity[19] is a quantum mechanical phenomenon characterized by the complete absence of electrical resistance in certain materials when cooled below a critical temperature. Discovered in 1911 by Dutch physicist Heike Kamerlingh Onnes, this discovery marked a significant milestone in condensed matter physics and sparked decades of research aimed at understanding the underlying principles and potential applications of this remarkable state of matter. Superconductivity has played a central role in advancing technology, including applications in magnetic resonance imaging (MRI), particle accelerators, and quantum computing. The journey to uncover its microscopic origins, however, has been both complex and transformative for the field of theoretical physics.

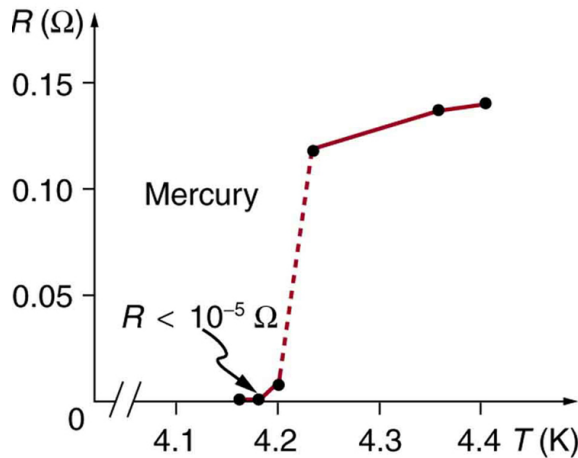


Figure 1.1: Experiment by H. K. Onnes on Mercury superconductivity transition

The phenomenon was first observed in 1911 when Onnes was investigating the properties of mercury at extremely low temperatures. By cooling mercury to just above absolute zero (4.2 K), he discovered that its electrical resistance suddenly dropped to zero. This meant that electric current could flow indefinitely without energy dissipation, a stark contrast to the behavior of normal conductors where resistance increases with decreasing temperature. Onnes' finding initially puzzled the scientific community, as it defied classical physics' understanding of how electrons behave in solids. It was only later that the quantum mechanical nature of superconductivity was revealed, revolutionizing the understanding of electrical conduction.

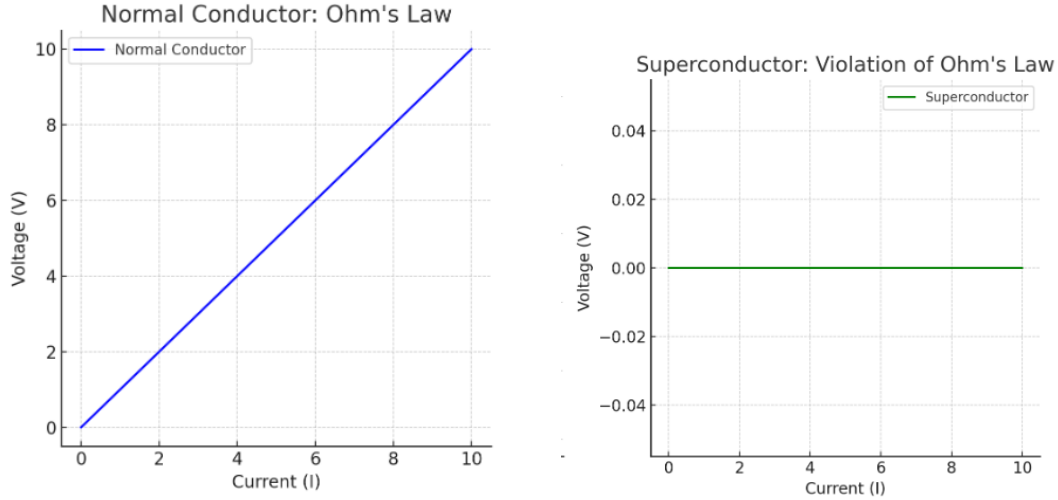
1.1.1 Violation of Ohm's Law

Superconductivity represents one of the most striking deviations from the classical laws of electromagnetism, particularly Ohm's Law[9].

Ohm's Law, one of the foundational principles in electrical theory, states that the current I passing through a conductor between two points is directly proportional to the voltage V across the two points, provided the temperature remains constant. Mathematically, this relationship is expressed as

$$V = IR$$

The relationship is graphically represented in Fig 1.2(a).



- (a) The voltage increases linearly with current, following Ohm's Law where $V = IR$ (b) In the superconducting state, the voltage remains zero regardless of the current

Figure 1.2: Resistance behavior in Conductors and Superconductors

Where V is the voltage, I is the current, and R is the resistance of the conductor. The resistance R , for conventional conductors like copper or aluminum, depends on the material's properties and temperature. As temperature decreases, the resistance typically decreases but never vanishes completely. This relationship forms the cornerstone of classical electrical theory. However, in superconductors, this relationship breaks down completely once the material is cooled below its critical temperature.

Superconductivity is a state of matter where electrical resistance drops to exactly zero below a certain critical temperature (T_c). This is a direct violation of Ohm's Law because, in the superconducting state, resistance R becomes zero, which means that for any finite current I , the voltage V across the material should also be zero according to Ohm's Law. In practical terms, a superconductor can carry an electrical current indefinitely without any applied voltage or energy loss.

Ohm's Law fundamentally assumes that electrical resistance is always present, even if it becomes very small. This resistance arises because of electron collisions with the atomic lattice and impurities in the material, which scatter the electrons and cause energy dissipation. In superconductors, this scattering mechanism ceases entirely once the critical temperature is reached, as electrons form Cooper pairs and condense into a coherent quantum state. In this state, the electron pairs can move through the crystal lattice without scattering, leading to a perfect flow of current. Therefore, Ohm's Law, which was developed for resistive materials, cannot apply to superconductors because they do

not exhibit resistance in this regime.

To understand why superconductors do not follow Ohm's Law, we must first look at the underlying mechanics that cause resistance in normal conductors. In a normal metal, resistance arises from interactions between the flowing electrons and the ions of the atomic lattice. As the electrons move through the lattice, they scatter off the vibrating atoms (phonons) or impurities, which leads to energy dissipation in the form of heat. This dissipation is described by Joule's Law, which states that the power P lost to heat in a conductor is proportional to the square of the current I and the resistance R

$$V = I^2 R$$

In superconductors, however, once the temperature drops below the critical threshold, the electrons form pairs (known as Cooper pairs) due to attractive interactions mediated by lattice vibrations (phonons). Cooper pairs behave collectively as a Bose-Einstein condensate, where all pairs occupy the same quantum state. In this state, the Cooper pairs move through the lattice without scattering, resulting in zero resistance. As a result, the current can flow without any voltage drop or power dissipation, effectively rendering Ohm's Law invalid in the superconducting phase.

1.1.2 Properties and Application of Superconductor

Superconductors exhibit a range of remarkable properties that distinguish them from normal conductive materials.

Zero Electrical Resistance: The most defining and remarkable property of superconductors is their ability to exhibit zero electrical resistance when cooled below a certain critical temperature. This phenomenon, which distinguishes superconductors from ordinary conductive materials, allows the uninterrupted flow of electric current without any dissipation of energy as heat. In conventional conductors, such as copper or aluminum, resistance arises due to the scattering of charge carriers (electrons) by impurities, phonons (lattice vibrations), and defects. However, in superconductors, these scattering mechanisms are suppressed entirely below critical temperature, leading to a state where resistance vanishes completely.

The zero-resistance behavior of superconductors can be explained through the microscopic theory of superconductivity, known as the Bardeen-Cooper-Schrieffer (BCS) theory[11]. According to this theory, below the critical temperature, electrons near the Fermi surface form bound pairs, termed Cooper pairs, due to an attractive interaction mediated by phonons. These Cooper pairs condense into a coherent quantum state that extends over the entire material, resulting in a collective behavior of the electrons. Unlike in normal metals, where individual electrons scatter off impurities or lattice vibrations, the Cooper pairs move without resistance because their quantum wavefunctions are coherent and do not suffer from the same scattering processes. The BCS theory predicts that the formation of Cooper pairs leads to the emergence of an energy gap (Δ) in the electronic density of states, below which no single-particle excitations can occur. This energy gap suppresses electron scattering, as the thermal energy at low temperatures is insufficient to break the Cooper pairs and create excitations. As a consequence, the electrical resistivity of the material drops to zero when the temperature is below, which can be expressed as

$$\rho(T) = 0 \quad T < T_c \tag{1.1}$$

Where $\rho(T)$ is the electrical resistivity as a function of temperature, and T_c is the critical temperature.

Meissner Effect: The Meissner Effect, discovered in 1933 by Walther Meissner and Robert Ochsenfeld[17], is a defining feature of superconductivity. It refers to the complete expulsion of magnetic fields from the interior of a superconductor when it transitions below a critical temperature to the superconducting state.

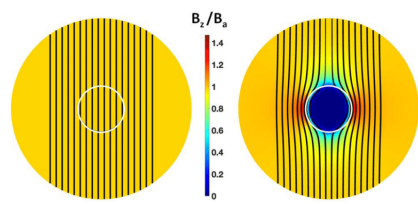


Figure 1.3: Illustration[4] of Meissner effect

This behavior contrasts with that of ordinary conductors, which allow magnetic fields to penetrate them partially, even in their most conductive states. When a material is cooled below critical temperature, it not only exhibits zero electrical resistance but also undergoes a dramatic shift in its electromagnetic properties. The Meissner Effect ensures that magnetic flux is expelled from the bulk of the superconductor, rendering the magnetic field within the material effectively zero. This phenomenon implies that superconductivity is not merely a state of perfect electrical conduction, but also a quantum state that actively resists magnetic field penetration. Mathematically, the Meissner Effect can be described by the London equations[15], developed by Fritz and Heinz London in 1935. These equations explain how superconductors shield their interior from external magnetic fields by generating surface currents that oppose the applied magnetic field. The magnetic field $B(x)$ inside a superconductor decays exponentially from the surface into the material over a characteristic length known as the London penetration depth λ

$$B(x) = B_0 \exp \frac{x}{\lambda} \quad (1.2)$$

where B_0 is the external magnetic field and x is the depth into the superconductor. This shows that the magnetic field quickly diminishes as it moves inward, confining the magnetic flux to the surface layer. The Meissner Effect is a key feature distinguishing superconductors from perfect conductors. A perfect conductor, while having zero electrical resistance, would allow magnetic fields to become "frozen" inside the material when it is cooled below T_c . In contrast, a superconductor actively expels pre-existing magnetic fields when transitioning into the superconducting state, showing that superconductivity involves more than just the elimination of resistance; it represents a distinct quantum phase. This expulsion of magnetic fields also underpins the phenomenon of magnetic levitation, where superconductors can repel magnets and levitate above them due to the repulsive forces generated by the Meissner Effect.

The Meissner Effect demonstrates how a superconductor when cooled below its critical temperature (often using liquid nitrogen), expels magnetic fields from its interior. This causes the magnet to levitate above the superconductor, as shown in Fig 1.4. The foam container helps to maintain the low temperature needed for this effect to occur. This illustration provides a clear, simplified view of the experimental setup used to demonstrate this fascinating property of superconductors. The image effectively illustrates the Meissner Effect, showing a magnet levitating above a superconductor cooled by liquid nitrogen in a foam container. This setup demonstrates the unique magnetic field expulsion property of superconductors.

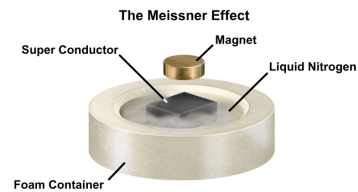


Figure 1.4: Meissner effect demonstrate magnet levitation at room temperature [13]

Persistent Currents in Superconductors: When an electrical current is initiated, it can flow indefinitely without experiencing any energy dissipation. This behavior is attributed to the formation of Cooper pairs—bound states of electrons with opposite spin and momentum—which condense into a collective ground state. The absence of electrical resistance is thus a direct consequence of the Cooper pairs moving coherently through the lattice without scattering off impurities or lattice vibrations. The concept of persistent currents can be elaborated through the framework of the London equations, which describe the electromagnetic properties of superconductors. According to the London theory, a supercurrent[12], once established, remains unchanged as long as the superconducting state is preserved. This phenomenon arises because the supercurrent is not associated with an electric field; rather, it is driven by the phase gradient of the superconducting wave function. Consequently, the current maintains its flow indefinitely, a clear deviation from conventional conductive materials where resistance causes a gradual decay of the current.

Furthermore, the phenomenon of persistent currents is profoundly demonstrated in superconducting rings and loops. When a superconducting ring is subjected to a magnetic flux, the resulting current establishes a quantized magnetic flux pattern, as described by the Aharonov-Bohm effect. This quantization is linked to the fundamental principle of flux quantization in superconductors, which asserts that the magnetic flux through a superconducting loop is quantized in units of the flux quantum (Φ_0). The persistent currents in these rings, therefore, represent a macroscopic quantum effect, where the current persists without attenuation. The concept of persistent currents is harnessed in technologies such as superconducting magnets and quantum computing. Superconducting magnets exploit the absence of resistive losses to generate extremely strong and stable magnetic fields, crucial for applications in medical imaging and particle accelerators. In quantum computing, persistent currents in superconducting qubits offer the potential for creating stable quantum states with minimal decoherence.

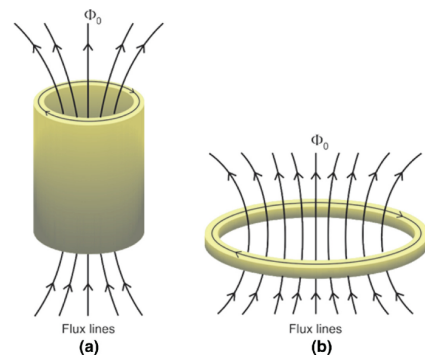


Figure 1.5: Persistent Current Effects in Superconductors

1.1.3 High temperature Superconductors

The search for materials that exhibit superconductivity at higher temperatures led to the discovery of high-temperature superconductors (HTS), which represent a significant breakthrough in condensed matter physics. These materials display superconductivity at temperatures far above those of conventional superconductors, challenging existing theoretical frameworks and promising wide-ranging technological applications. The discovery of HTS was made by Johannes Georg Bednorz and Karl Alexander Müller in 1986 [5], who identified superconductivity in a copper-oxide perovskite material at temperatures as high as 35 K, a record at the time. This discovery sparked significant interest in the field and led to a series of advancements, including the identification of yttrium barium copper oxide (YBCO), a material with a critical temperature of 93 K, high enough for cooling with liquid nitrogen. This marked a substantial leap forward, as the use of liquid nitrogen is considerably more practical and cost-effective than liquid helium, which is required for cooling conventional superconductors.

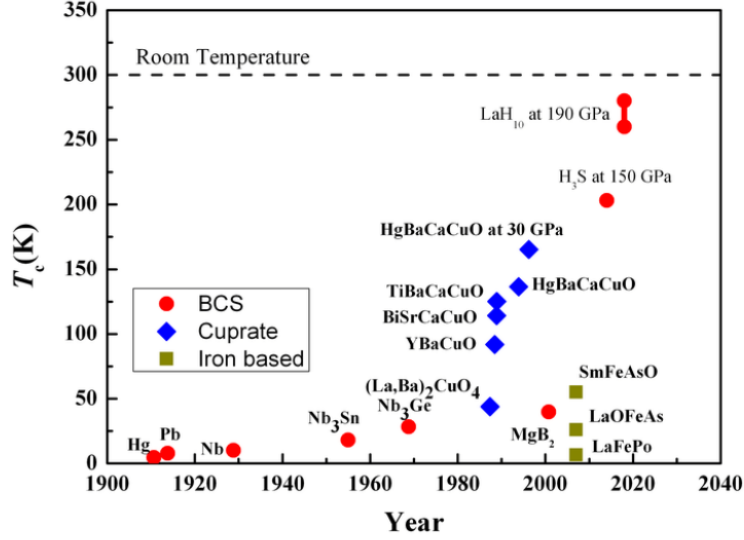


Figure 1.6: Timeline of discoveries of superconductors with critical temperature

1.2 Superconductivity with Disorder

In condensed matter physics, disorder refers to the irregularities or imperfections within the atomic or electronic structure of materials. Unlike perfectly ordered crystalline lattices, real-world materials contain various forms of disorder such as point defects that occur when atoms are missing from or displaced within the lattice. Impurities is define as Atoms of different elements that can be introduced into the material either intentionally (for doping) or unintentionally, altering the local electronic environment. Structural defect include grain boundaries, dislocations, and stacking faults, which are deviations from the regular atomic arrangement. And topological defect is an imperfection in the long-range order of a system that can affect the electronic states. Disorder in materials typically leads to phenomena such as scattering of electrons, reduced mobility, and sometimes localized states. In conventional conductors, this scattering increases electrical resistance, and in extreme cases, can lead to electron localization, where electrons become trapped in spatially localized states and cease to contribute to conduction (Anderson localization). However, superconductors are unique in that they can exhibit surprising resilience to certain types of disorder while being sensitive to others, depending on the nature of the superconducting pairing and the type of disorder present.

One of the earliest theoretical breakthroughs in understanding the effect of disorder on superconductors came from Philip W. Anderson in the late 1950s. The Anderson theorem[3] established that non-magnetic disorder has little to no effect on the superconducting transition temperature in conventional s-wave superconductors, where the pairing of electrons (Cooper pairs) occurs in a state with zero orbital angular momentum. This resilience arises from the fact that superconductivity is a macroscopic quantum state. While disorder scatters individual electrons, it does not significantly affect the coherent pairing of electrons into Cooper pairs as long as the disorder remains non-magnetic. The energy gap associated with Cooper pair formation remains intact, meaning that transition temperature and the overall superconducting properties are preserved. However, the Anderson theorem holds under specific conditions: disorder must be sufficiently weak, and the scattering must remain

non-magnetic. Magnetic impurities, on the other hand, break time-reversal symmetry and introduce pair-breaking effects that can significantly suppress superconductivity.

When magnetic impurities are introduced into a superconductor, they can have a devastating impact on superconductivity. Unlike non-magnetic disorder, magnetic impurities interact with the electron spin, thereby breaking the time-reversal symmetry essential for the formation of Cooper pairs. This leads to the suppression of the superconducting gap and the reduction of the critical temperature. In extreme cases, magnetic disorder can destroy the superconducting state, as described by the Abrikosov-Gor'kov[2] theory.

The pair-breaking mechanism is linked to the fact that Cooper pairs in conventional superconductors consist of two electrons with opposite spins and momenta. Magnetic impurities locally disturb the spin environment, causing the dephasing of the electron pairs, thereby weakening or breaking the superconducting condensate. In this context, the superconducting transition temperature is reduced, and the material may transition into a non-superconducting phase.

1.3 Purpose of Study

This study investigates the relationship between superconductivity and disorder, with a specific focus on how randomness in superconductors affects the critical temperature(t_c). The primary aim is to determine the impact of disorder on the superconducting transition temperature. By examining the behavior of the critical temperature and identifying the key variables it depends on, this research seeks to validate the hypothesis that controlled disorder can have a positive influence on superconductivity.

1.3.1 Research Questions

Solve the stochastic equation numerically and generate a dataset that incorporates the critical temperature of superconductors in both disordered and ideal (disorder-free) conditions. Perform data analysis to analyze the influences of disorder on the critical temperature of superconductors.

1.3.2 Study Objectives

The first objective is to establish a relationship between the critical temperature and the coupling constant or Cooper constant(λ) through a mathematical stochastic equation. This case will be treated as the ideal scenario, without any randomness introduced. Introduce disorder in the Cooper constant(λ), modeled as a random variable drawn from a normal distribution $\mathcal{N}(\mu, \sigma^2)$. For each random case, the mean critical temperature will be calculated, producing a dataset generated entirely from the mathematical model. Conduct preliminary data analysis to observe and understand the behavior of the critical temperature in response to varying levels of disorder. This analysis will offer insights into how randomness impacts superconductivity. Finally, four machine learning models—Random Forest, Gaussian Process Regression (GPR), Monte Carlo with Linear Regression, and Variational Autoencoder (VAE)—will be applied to further investigate the stochastic nature of superconductivity. These models will provide quantitative evidence to assess the effect of disorder on the critical temperature, helping to conclude the role of randomness in enhancing or suppressing superconductivity.

Chapter 2

Literature Review

2.1 Background Study

In 1911, Heike Kamerlingh Onnes made a pivotal discovery regarding the behavior of mercury's electrical resistance when cooled to extremely low temperatures, observing that resistance vanished entirely at a certain threshold temperature, a phenomenon later termed superconductivity [19] using liquid helium to cool mercury at temperatures as low as 4.2K (-268.95°). He observed that there was a sudden and complete disappearance of electrical resistance. The resistance dropped to zero which indicated that an electric current was flowing without any loss of energy. Onnes concluded from his remarkable experiment that mercury has entered into a new state of matter, termed "Superconductivity". Later in 1913, Onnes's work earned him the Nobel Prize. The In-depth study of superconductivity on a microscopic level and its properties came under study in 1957 by John Bardeen et al. and termed BCS theory. The theory, developed by John Bardeen and his collaborators, focused on several key aspects of superconductivity, including the formation of Cooper pairs, the energy gap, and the critical temperature [11]. This work won him and his team the Nobel Prize in Physics in 1972.

The study of impurities in superconductors has long been a subject of significant interest in condensed matter physics particularly following the seminal work of Ki-ichi Nakamura et al.[18] in 1959 where he investigated the influence of non-magnetic properties on superconductors based on Bardeen, Cooper, and Schrieffer's (BCS) theory. It has been observed that impurities scattered over the electrons lead to reduced electron-electron interaction and reduced superconductivity transition temperature. In the same year, P.W. Anderson et al.[3] developed a theory addressing how disorder, particularly impurities, affects the superconducting property of the material. His work laid the theoretical foundation for understanding how non-magnetic impurities do not significantly affect the critical temperature. However, this result has faced significant contradictions and refinements over time, particularly with the advent of more detailed theoretical and experimental studies. The same year, A.A.Abrikosov et al.[2] studied the paramagnetic impurities on superconductors which produce a substantial amount of disturbance on superconductivity, unlike the non-magnetic properties. A.A.Abrikosov and Gorkov observed that magnetic impurities create a localized magnetic movement that disrupts the time-reversal symmetry of the Cooper pair and results in the reduction of the energy gap and a decrease in critical temperature. The idea originated from the work of Heinz London et al.(1937)[15] observation that impurities could affect superconductivity might stem from the general understanding at the time that real-world materials would exhibit deviations from ideal superconducting behavior. This laid the groundwork for understanding how disorder can affect

superconductivity with magnetic and non-magnetic impurities. For broader knowledge, G.Boato et al.(1966) explore how the transition metal impurities affect the critical temperature of various conducting elements. The outcome of the decrease in critical temperature was consistent across the material tested(Aluminum (Al), Zinc (Zn), Indium (In), and Tin (Sn)). These findings were consistent with the general expectations of the time and contributed to the broader knowledge of impurity effects in superconductors. All the work mentioned indicates that disorder in superconductors only helps to decrease the critical temperature and this behavior is generally more conventional.

In 1986, IBM researchers J. Georg Bednorz and K. Alex Müller made a significant breakthrough by discovering high-temperature superconductors, earning them the Nobel Prize in Physics the following year [5]. This considerable breakthrough marked the first successful demonstration of superconductivity in ceramic materials. In such material, increasing the disorder with the help of doping or another form of structure imperfection may sometimes increase its critical temperature. These phenomena occur due to the complex interplay between electrons correlation or lattice structure. In certain cases, disorder can elevate the density of states at the Fermi level, which may lead to a rise in the critical temperature [1]. M.P.A. Fisher and colleagues explored this concept in 1990, focusing on two-dimensional systems. Their research demonstrated that disorder-induced localization effects could increase the density of states at the Fermi level, resulting in higher superconducting critical temperatures [8]. This research laid the groundwork for further studies on the complex relationship between disorder and superconductivity in 2D systems, showing that disorder does not always suppress superconductivity but can enhance it under the right conditions. To solidify this phenomenon, D. Kowal experimented with two-dimensional systems showing that disorder can enhance superconductivity in a thin film of indium oxide and observed that as disorder increased, the superconducting transition temperature also increased, providing strong evidence for the disorder-driven enhancement of critical temperature. These studies advocate that the appropriate amount of disorder in the superconductors can help to increase critical temperature. The property of materials plays a pivotal role in finding the proper amount of disorder.

2.2 Related Work

Machine learning techniques and tools were integrated into every sector of the physical world, including the study of superconductivity and its relationship with disorder. In 2017, Valentin Stanev et al.[21] was one of the first to implement machine learning (ML) techniques in superconductivity. Valentin’s study focused on creating machine learning models to estimate the critical temperature of superconducting materials. The research utilized Random Forests[10], Neural Networks, and Support Vector Machines[7] to identify important factors like chemical composition and crystal structure, leveraging these features to predict novel superconducting materials. This integration of machine learning techniques was followed by researchers and in 2019, S.R.Xie [23] explored the application of machine learning to predict the superconducting critical temperature of various materials using Gaussian Process Regression(GPR)[20] as a primary approach. The approach allowed the authors to model the nonlinear relationship between material properties and critical temperature, making it a valuable tool for guiding future experimental efforts in the search for new superconductors. A similar study was conducted by Anton Matasov et al.[16] in 2020 where the primary goal of the study was to develop a computational model using regression techniques that can predict the critical temperature of known superconductors and identify new materials that might exhibit superconductivity at high temperatures. All the studies using machine learning models focused on the prediction of critical temperatures or finding new material that exhibits the same properties as superconductors. In the same year, a more precise approach was explored by Thanh Dung Le et

al.[14] Using the variation Bayesian neural network(VBNN) approach. Unlike the neural network approach, VBNN incorporates probabilistic reasoning to handle uncertainty, which is crucial in cases with limited or noisy data. The ability to predict the critical temperature more precisely but also provide uncertainty estimates for each prediction, making the model more robust. VBNNs provide more advances in enhancing the discovery of new superconductors, especially at high temperatures. The use of Bayesian techniques offers an advantage over deterministic methods by allowing better generalization and reliable predictions in the domain of superconductivity. In 2020, Xhang and Xu[24] conducted a study to predict the critical temperature of doped superconductors using the Gaussian Process Regression(GPR) technique. GPR is well-suited to handle uncertainties and non-linear relationships between the lattice parameters and critical temperature. GPR also provides both predictions and confidence intervals, allowing for an understanding of the uncertainty in predictions and providing high accuracy critical temperature as a result. In recent times, more advanced machine learning has been used by Dmitry Viatkin et al.[22] using convolution neural networks technique. The convolutional neural networks (CNN) approach was used to predict the critical temperature of superconducting materials by analyzing their chemical formulas. In a typical scenario, CNN used for image processing, is adapted in this case to process and learn patterns from chemical formulas. The model extracts relevant features from the chemical structure and composition of superconductors, learning the relationships between these features and their corresponding critical temperature.

2.3 Gap in the Study

In previous studies, the primary focus of machine learning techniques has been discovering new superconducting materials, particularly those exhibiting high critical temperatures. These efforts have predominantly aimed to deepen our understanding of the mechanisms driving superconductivity at elevated temperatures. In contrast, the role of disorder and its influence on critical temperature have often been relegated to secondary studies. In recent years, numerous studies have employed machine learning models to predict the critical temperatures of superconductors. However, these studies have largely overlooked the role of disorder in superconducting materials. Disorder has often been treated as noise in superconductivity research rather than as a significant parameter influencing critical temperature. Consequently, the majority of research has not considered disorder as a variable in predictive models. The datasets utilized in these studies typically comprise properties of real superconducting materials, focusing primarily on intrinsic characteristics such as lattice parameters or chemical composition. Furthermore, the machine learning approaches employed in prior research have generally been limited to one or two techniques, concentrating on predicting the critical temperatures of specific superconductors. This narrow focus has resulted in models that are specialized for certain materials rather than providing a broader understanding applicable to general superconductors. This gap highlights the need for machine learning models that incorporate the effects of disorder and superconductivity. Employing multiple machine learning models on datasets generated from mathematical stochastic equations can offer a more comprehensive approach to understanding the relationship between superconductivity and disorder. Researchers can better capture the complex interactions between Cooper constant and critical temperature by integrating multiple machine learning techniques.

Chapter 3

Data Generation

A superconductor's property can be expressed in 2 different forms: with and without the presence of randomness. Randomness is defined as the frequent changes that occur in the properties of the material on the Fermi level where electrons will pair up to form a Cooper pair due to an attractive interaction mediated by lattice vibration(phonons) due to the presence of impurities in a superconductor. This affects the change of critical temperature (t_c), critical magnetic field, electrical resistivity, gap suppression, and vortex dynamics[6] etc. In this study, we have only focused on the relationship between superconductivity, cooper constant(λ), critical temperature(t_c), and energy of an electron in superconducting state(ϵ) which is expressed in derived stochastic equation form.

$$\lambda \int_0^{\frac{1}{t_c}} \tanh.\epsilon. \frac{d\epsilon}{\epsilon} = 1 \quad (3.1)$$

This equation has been solved numerically as a function of critical temperature and cooper pair constant variable $f(t_c, \lambda) = 1$ using SciPy libraries and implemented in Python language. Due to a non-linear function of tangent hyperbolic, it introduces complexity into the integral, and therefore mathematical solutions have not been approached. The solution will be used for the ideal superconductor which contains no randomness.

The given stochastic equation behaves randomly only if there is randomness in the cooper pair(λ). Let's suppose that coopers pair as random variables that are generated by the normal distribution function and can be expressed as:

$$f(\lambda) = \frac{1}{\sqrt{2\pi}\sigma} . e^{-\frac{(\lambda-\lambda_0)^2}{2\sigma^2}} \quad (3.2)$$

The normal distribution function of the cooper pair is defined. The variance throughout the whole study has been constant and considered $\sigma = 0.1$. Cooper pair with no randomness in nature will be considered as the ideal cooper pair λ_0 in a random instance.

For every random state value of the coopers pair, co-responding critical temperature have been calculated. The data point generated for random state cooper pair filtered out within the range of [0, 0.5] because the range is more conventional for the majority of superconductors and to avoid instability that could disrupt the superconducting state. Now, the mean value of critical temperature(t_c) will be drawn as a function of Cooper constant(λ_0).

We have generated 1200 data points for critical temperature for every cooper pair random state value and saved them into an Excel file for analysis purposes.

The derived equation is stochastic due to the randomness and unpredictability of coopers pair formation and it highly affects change in critical temperature(t_c). Analytical solutions of the superconductivity and disorder relation equation help to find the data points within the viable range of cooper pairs that exist for the majority of superconductors. To precisely capture the relationship between critical temperature and cooper pair in a random state for the superconductors, a stochastic machine learning model is required. To establish such a relation with the suggested parameters, we have applied 4 viable models.

The relation between the Cooper pair and critical temperature will be drawn for 2 scenarios where randomness is present within the equation and without the presence of any randomness.

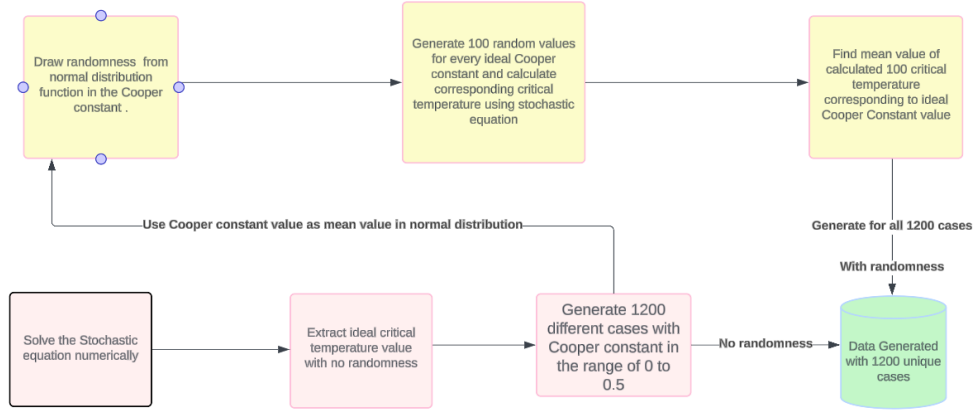


Figure 3.1: Flow diagram for the generation of dataset

3.1 Superconductivity without randomness

Superconductors do not consist of any type of impurities within and behave such that there is no disorder caused by to cooper pair of electrons and every electron moves streamlined. Such conditions are not pragmatic in the real world but to find an ideal relationship between cooper pair(λ) and critical temperature(t_c). To establish the relation, we have solved the given stochastic equation analytically with the help of the SciPy library and defined the range for the coopers pair as well $\lambda \in [0, 0.5]$. The relationship is expressed using graphical visualization.

To ensure that the value of critical temperature is calculated within the range of the cooper pair value, a few random instances have been displayed in Table 1.

Coopers pair value(λ)	Critical temperature(t_c)
0.068333	0.000001
0.155538	0.003657
0.242744	0.036852
0.329949	0.109472
0.417155	0.206125

Table 3.1: Critical temperature calculated for random coopers pair value within the defined range

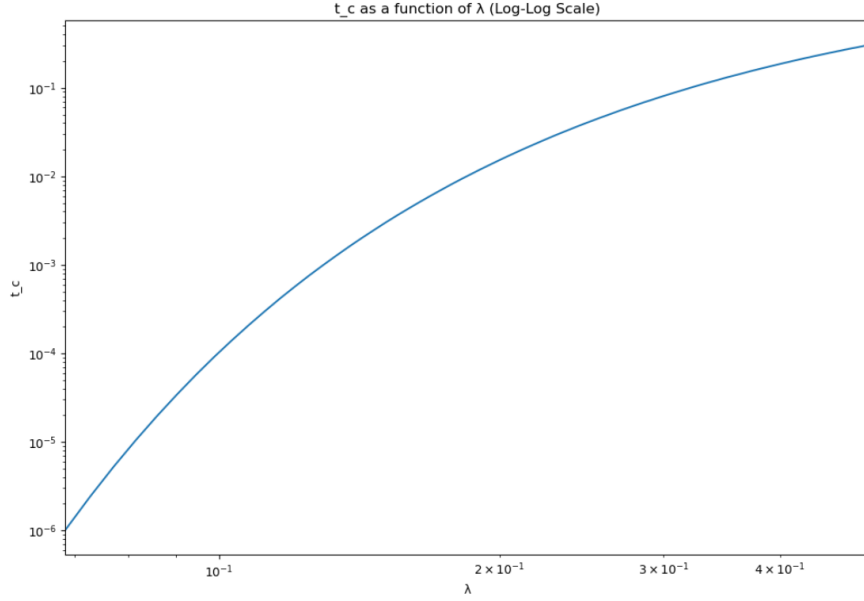


Figure 3.2: Critical temperature t_c as function of coopers pair (λ) consisting no randomness

3.2 Minimum Value of Cooper Pair(λ)

The derived stochastic equation has solutions that exist within the range of where $\lambda \in [0, 0.5]$. To find the minimum value at which the critical temperature can be calculated, we are required to calculate the maximum value for the integral. This is implemented because when the value of cooper pair(λ) is considered, it produces no result, and therefore, values for critical temperature can not be calculated for such instances. Therefore, we are required to update the range for the cooper pair.

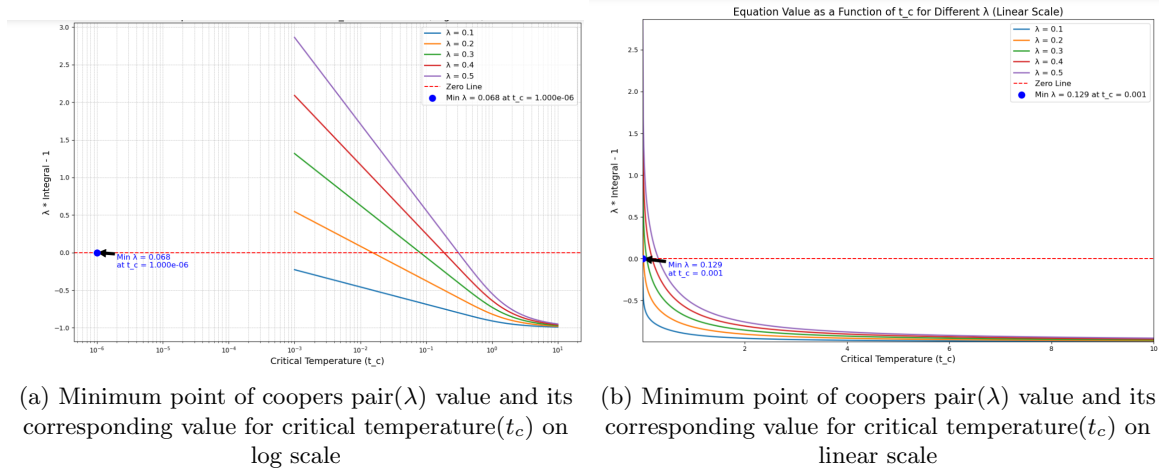


Figure 3.3: Minimum points for parameters and behavior of critical temperature at different points

The relationship derived from the equation is such that it depends on the upper limit of integral $\frac{1}{t_c}$. In Figure 3.2(a), we can visualize the minimum point for coupling constant(λ) and corresponding critical temperature(t_c). The maximum value of the integral is 14.634291 occurs when the critical temperature is at its lowest with a value of 0.000001 and a corresponding value of coupling constant is 0.06833 using a log scale visualization. In Figure 3.2(b), linear scale visualization is used and the maximum value for integral is 7.726535 occurs when the critical temperature is 0.001 and the corresponding coupling constant value is 0.129.

Since λ is inversely related to the integral, the maximum value of the integral corresponds to the minimum value of λ . This indicates that the system requires less coupling strength(λ) to maintain superconductivity as more energy states contribute to the superconducting phase at lower critical temperatures. As the critical temperature t_c decreases, the upper limit of the integral, $\frac{1}{t_c}$, increases. This leads to an expansion in the range of integration, allowing more energy states ϵ to contribute to the superconducting phase. The maximum value of the integral occurs at the minimum critical temperature t_c because, at lower temperatures, the expanded integration range allows more energy states to contribute to the superconducting phase. This leads to a larger integral value corresponding to a minimum value of λ .

The logarithmic scale highlights small values of t_c and thus finds the maximum integral at a very low critical temperature ($t_c = 0.000001$). This results in a lower minimum $\lambda = 0.068333$, which is within the defined range of 0 to 0.5. The log scale is useful for emphasizing behavior at extremely small t_c , which is relevant when studying superconductors operating at very low temperatures. In contrast, the linear scale emphasizes larger values of t_c , leading to a higher minimum $\lambda = 0.129424$ at a higher critical temperature ($t_c = 0.001000$). While this value is also within the defined range, the linear scale may not capture the full extent of behavior at very low t_c , which can be crucial for accurate modeling of superconductors. Considering the defined range of λ between 0 and 0.5 and the need to accurately capture behavior at very low t_c , the logarithmic scale provides a more comprehensive view of the coupling constant's range. It allows us to observe how λ behaves at extremely low critical temperatures, which is important for understanding superconductors in practical scenarios.

Therefore, while both scales are useful, the results from the logarithmic scale are more aligned with the expected physical range and behavior of superconductors, making it the preferred choice for determining the appropriate range of λ . The new coupling constant(λ) range is $\lambda \in [0.07, 0.5]$.

Note: The minimum value of the Cooper pair may vary depending on the system in which it is implemented. For instance, high-performance computational systems can yield more precise estimations due to their enhanced computational accuracy and processing capabilities.

3.3 Superconductivity with Disorder

In the real world, superconductors generally have disorders due to their impurities. This disorder causes the change critical temperature(t_c). The change occurrence can be positive or negative. The existence of random behavior is caused by the coupling constant(λ). Therefore, we assumed that the coupling constant in the disordered or random state has been drawn out from the normal distribution function. The equation can be expressed as:

$$\mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left\{-\frac{(\lambda - \lambda_0)^2}{2\sigma^2}\right\}$$

The mean Cooper pair value, denoted as λ_0 , for the disordered state is defined as the mean value taken within the range. The standard deviation, denoted as σ^2 , is assumed to be 0.1. The

Cooper pair value for the random state, λ , is sampled within the $\lambda \in [0.07, 0.5]$ range. We have generated values for λ for each λ_0 and calculated the corresponding value of critical temperatures t_c . To express this mathematically, each λ_0 has the following dependent parameter.

$$\begin{array}{ccc} \lambda_1 & \rightarrow & T_1 \\ \lambda_2 & \rightarrow & T_2 \\ \lambda_3 & \rightarrow & T_3 \\ | & | & | \\ | & | & | \\ \lambda_n & \rightarrow & T_n \end{array}$$

We have a set of critical temperatures for the given Cooper constant(λ_0).From here, the mean value for critical temperature can be calculated as such

$$\langle T \rangle = \frac{1}{N} \sum_{i=1}^N T_i = \langle T \rangle (\lambda_0)$$

This calculated critical temperature depends on the distribution of function, which means depends on the λ_0 . This procedure has been performed repeatedly for different λ_0 . The value for λ_0 will run from 0 to 0.5. The value of the Cooper constant is a small value parameter. For instance, let's suppose that mean Cooper constant $\lambda_0=0.20$, then the first 5 pairs of Cooper constant and critical temperature in disorder are

λ	Critical temperature(t_c)
0.357075	0.137827
0.156277	0.003772
0.404817	0.191769
0.500000	0.307027
0.125954	0.000808

Table 3.2: 5 pairs value calculated when $\lambda=0.20$

The values for λ are generated from normal distribution. Then random variable value will be used to calculate the critical temperature with the help of equation (1). The next step is to calculate the mean value of the critical temperatures. For given instance,

$$\text{Mean critical temperature} = 0.094495$$

$$\text{Correlation between } \lambda \text{ and } t_c = 0.9602$$

$$\text{Median of critical temperature} = 0.018520$$

$$\text{Standard deviation of critical temperature} = 0.018520$$

The relationship between them can also be visualized using a graphical representation.

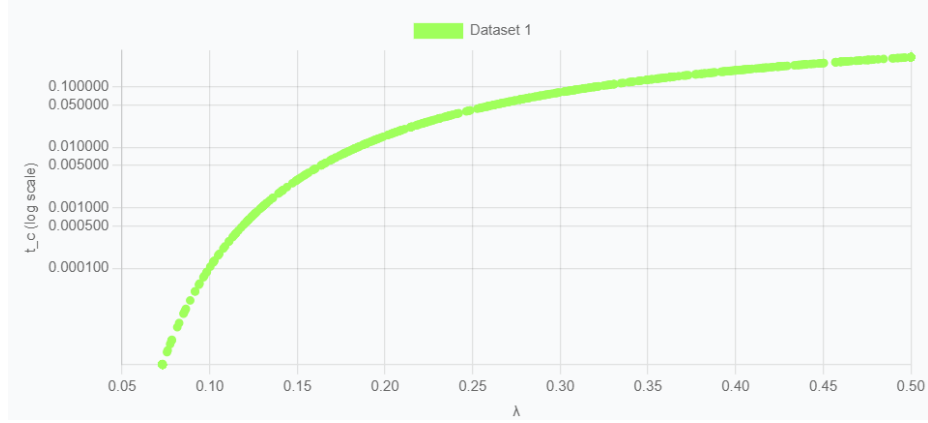


Figure 3.4: Critical temperature with randomness when $\lambda_0=0.2$

Using a similar method repeatedly, 1200 unique data points are generated. The sample of the dataset is demonstrated in Table 3.3

λ_0	λ_{mean}	$mean\ t_c$ (With disorder)	direct t_c (Without disorder)
0.49252	0.36384	0.18849	0.29781
0.38105	0.31552	0.15201	0.16438
0.10896	0.19871	0.05968	0.000234
0.22934	0.27291	0.11508	0.02896

Table 3.3: Sample from the generated dataset for critical temperature of superconductors

The dataset is generated for our randomness and without randomness in superconductivity. The parameters are λ_0 , the value within the defined range where $\lambda_0 \in [0, 0.5]$. The second parameter is λ_{mean} is generated using a normal distribution function, 100 unique values were generated and each unique value will have a corresponding critical temperature(t_c), the mean values of 100 such cases will give our third parameter which is $mean\ t_c$ (With disorder). And using λ_0 , when there is no randomness exists in the superconductivity, direct t_c (Without disorder) is calculated.

Chapter 4

Data Analysis

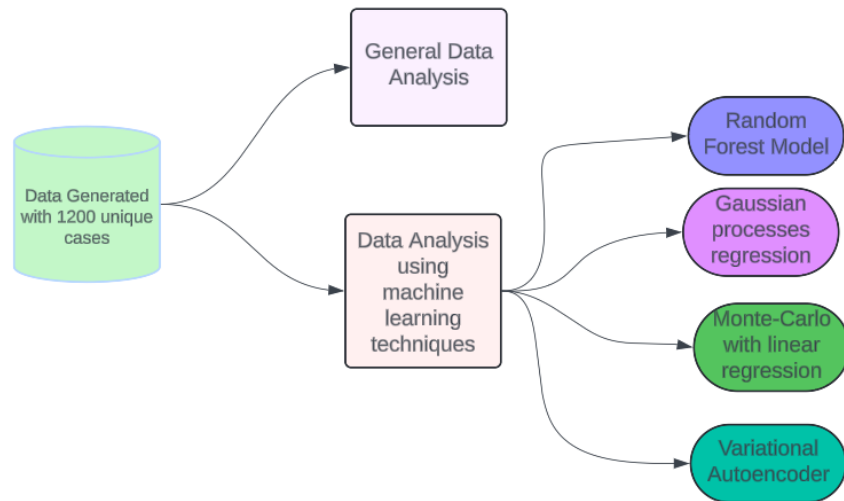


Figure 4.1: Flow diagram of Analysis

This chapter presents the analysis and implementations of machine learning models to investigate the behavior of critical temperature in superconductors in the presence of disorder. Machine learning techniques are particularly well-suited to capture the complex, non-linear relationships within the dataset, relationships that may be challenging to fully explore using traditional analytical methods. The generated dataset utilized for this analysis is clean and well-structured, containing no duplicate or missing values. Outlier detection has been deemed unnecessary because the calculated parameter variations in critical temperatures arise from the inherent randomness in the system, which is part of the expected physical dynamics. Removing these variations could obscure important insights since the coupling constant values are already confined within a predefined range. This study focuses on analyzing how randomness influences the critical temperature; therefore, large deviations are not

considered errors but manifestations of the variability introduced by the stochastic nature of the system.

Normalization and feature engineering are not required in this context. The data is directly interpretable without scaling or transformation, as the variables involved—particularly the critical temperature and coupling constants—are already in meaningful physical units. Since the objective is to explore the effects of randomness, introducing additional features or transformations would not enhance the analysis and may introduce unnecessary complexity.

4.1 General Analysis

In general analysis, we analyze the data without applying any machine learning model. This is implemented to learn about our dataset generated from mathematical equations to understand the relationship between coupling constant values and critical temperatures.

The ideal case in superconductivity is achieved when no randomness/disorder occurs in our superconductor, which implies that the superconductor contains no impurities.

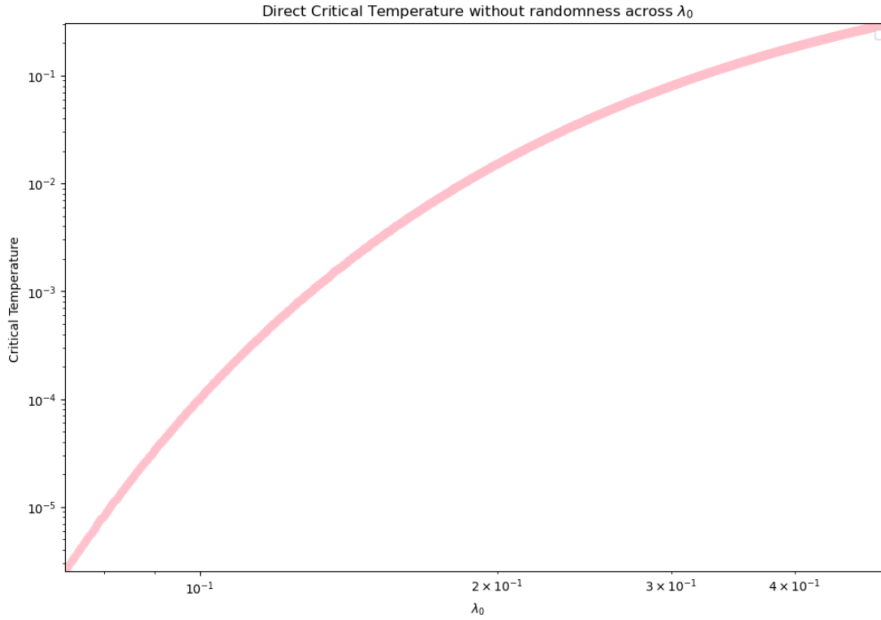


Figure 4.2: Superconductivity without randomness using log-scale visualization

From the figure, we can interpret that the critical temperature increases as coupling constant λ_0 increases. For better visualization, a log scale is used. The scattered plot is extremely streamlined and behaves like a line plot. The critical temperature without disorder spans a range from 0.000002 to 0.307. The correlation coefficient is found to be 0.96, which implies a strong positive linear relationship between critical temperature and coupling constant value. This suggests that as the coupling constant increases, the critical temperature will tend to increase predictably.

Introducing a disorder in superconductivity, the behavior of critical temperature will not be predictable for the same coupling constant values.

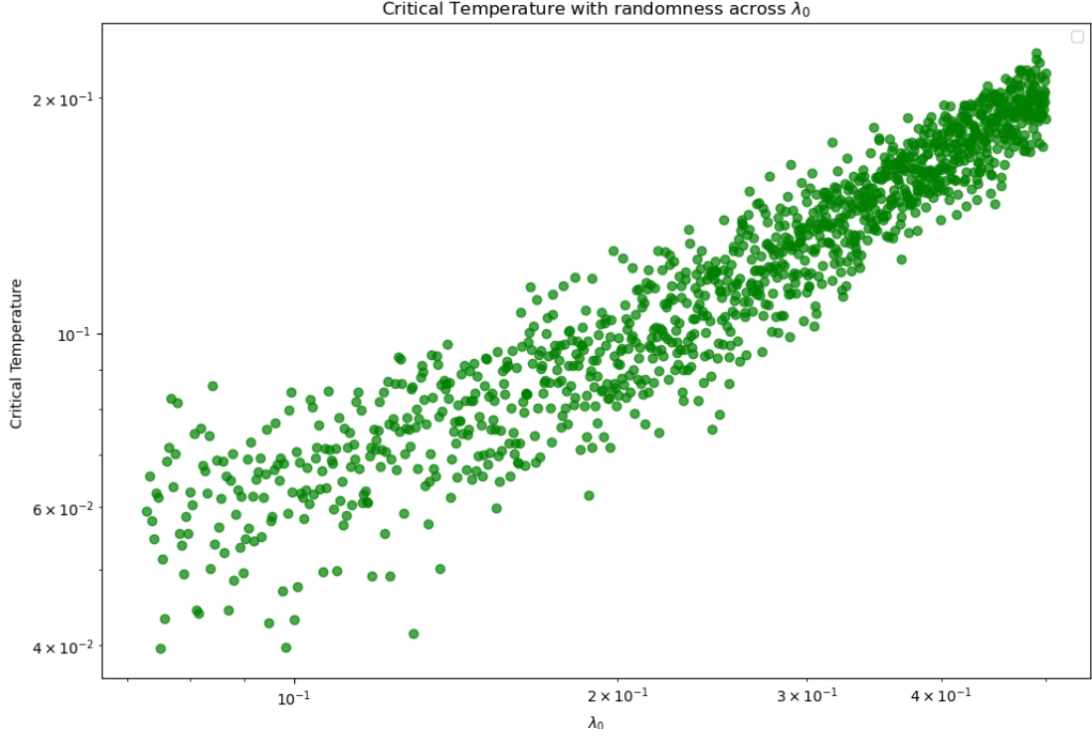


Figure 4.3: Superconductivity with disorder

The distribution of critical temperature exhibits significant variability across the range of values for the Cooper constant (λ_0). Notably, the scale of the critical temperature on the Y-axis varies considerably even for identical values of λ_0 . The critical temperature with disorder spans a range from 0.03968 to 0.2283. The observed correlation coefficient of 0.95 indicates a robust positive correlation, suggesting that as the Cooper constant increases, the critical temperature also tends to rise.

To make a comparison of critical temperature with random variables and without randomness, scattered plot visualization with linear scale is pursued. This will help us to understand the behavior of critical temperature(t_c) on the same scale shown in Fig 4.3. There is a significantly high critical temperature noted for the superconductor which carries a small Coupling constant(λ_0) value in the presence of randomness. This phenomenon can be attributed to the influence of the random process, which introduces occasional larger values of the Coupling constant using a normal distribution function. These larger values disproportionately elevate the mean critical temperature due to their significant impact on the average calculation. Consequently, the variability introduced by randomness at lower coupling constant values results in higher observed critical temperatures. Conversely, at larger λ_0 values, the effect of randomness diminishes. This reduction can be attributed to a potential saturation effect in the relationship between λ and T_c , where the influence of randomness becomes less pronounced. As λ_0 increases, the system's response to random fluctuations becomes less impactful, leading to a convergence of critical temperature with disorder and without disorder. This saturation suggests that the critical temperature becomes relatively stable and less sensitive to random variations at higher λ_0 , highlighting the complex interplay between randomness and system behavior at different scales of λ_0 .

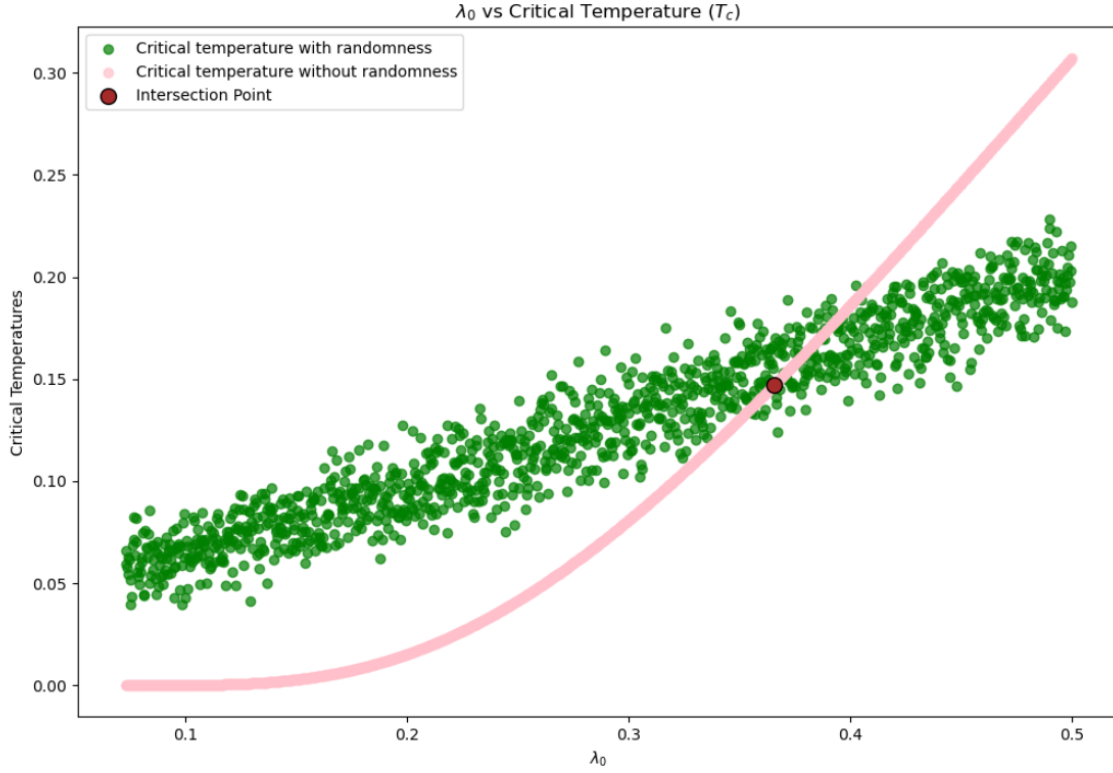


Figure 4.4: Superconductivity with disorder and superconductivity without disorder using linear scaling

To enhance the reliability of our study, we identified a critical intersection point where the critical temperatures of the superconductor, both with and without the influence of randomness, converge. The intersection point represents a significant reference where the impact of randomness on the critical temperature is balanced by the deterministic component. The coordinates of this intersection point are as follows::

$$\lambda_0 : 0.3657389491242702$$

$$\text{Critical temperature with disorder} : 0.1472484199055156$$

$$\text{Critical temperature without disorder} : 0.1472813914751555$$

The critical temperatures with and without the influence of randomness are nearly equal, with a difference of only 0.0001. This intersection point signifies a state where the two methods of calculating critical temperatures converge, indicating a point of equivalence in the behavior of the system under study.

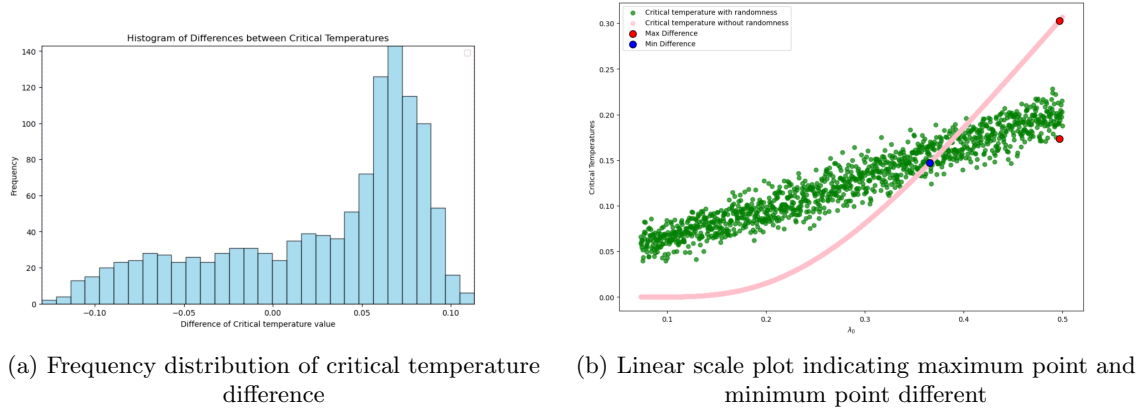


Figure 4.5: Maximum-minimum datapoint and frequency of critical temperature differences

From Figure 4.4, The analysis of the critical temperatures with and without randomness, as a function of λ_0 , revealed key differences in the behavior of these two systems. At $\lambda_0 = 0.4968$, the maximum observed difference between the critical temperatures occurred. The critical temperature with randomness, $t_c^{\text{random}} = 0.1733$, while the critical temperature without randomness, $t_c^{\text{direct}} = 0.3031$, resulted in a significant deviation of 0.1298. This disparity highlights the pronounced impact that randomness introduces into the system, substantially lowering the critical temperature.

At $\lambda_0 = 0.1980$, the maximum observed difference between the critical temperatures was identified. The critical temperature with randomness, $t_c^{\text{random}} = 0.1276$, contrasted with the critical temperature without randomness, $t_c^{\text{direct}} = 0.0145$, resulting in a substantial deviation of 0.11310. This notable disparity underscores the pronounced effect of randomness on the system, where the inclusion of randomness significantly elevates the critical temperature compared to the scenario without randomness. For visualization, a linear scale provides a more intuitive understanding of the differences in critical temperatures. These findings underscore the complex relationship between λ_0 and the critical temperature, with randomness playing a crucial role in determining the system's phase transition properties.

4.2 Machine learning Models

4.2.1 Random Forest Model

This method, pioneered by Leo Breiman in the early 2000s, aggregates multiple decision trees to improve both predictive accuracy and robustness. Each decision tree in the forest is constructed using a bootstrapped sample of the training data, and the final prediction is derived by averaging or majority voting across all trees. This ensemble approach significantly mitigates overfitting and enhances the generalization capability of the model. In the context of stochastic models, the RF model offers several advantages. Stochastic models, characterized by inherent randomness and uncertainty, require analytical approaches capable of accounting for these variabilities. The RF model's ability to handle high-dimensional data and complex feature interactions makes it well-suited for analyzing systems where randomness plays a crucial role. For instance, in studies where the coupling constant (λ_0) introduces variability in the critical temperatures of superconductors, the RF model can effectively disentangle the effects of randomness from the underlying signal. The RF model's robustness to overfitting makes it particularly useful when dealing with noisy or incomplete data,

which is common in stochastic simulations. By aggregating predictions from multiple trees, the RF model reduces the sensitivity to noise and ensures that the conclusions drawn about the impact of randomness are not unduly influenced by outliers or random fluctuations. To illustrate the utility of the Random Forest (RF) model in examining the impact of randomness on critical temperatures, our dataset has been generated from mathematical equations reflecting both random and deterministic states. In this setup, λ_0 is the stochastic variable influencing critical temperatures. The RF model is employed to analyze how variations in λ_0 affect critical temperatures, offering insights into the stochastic behavior of the system. By training the RF model on datasets representing different levels of randomness in λ_0 .

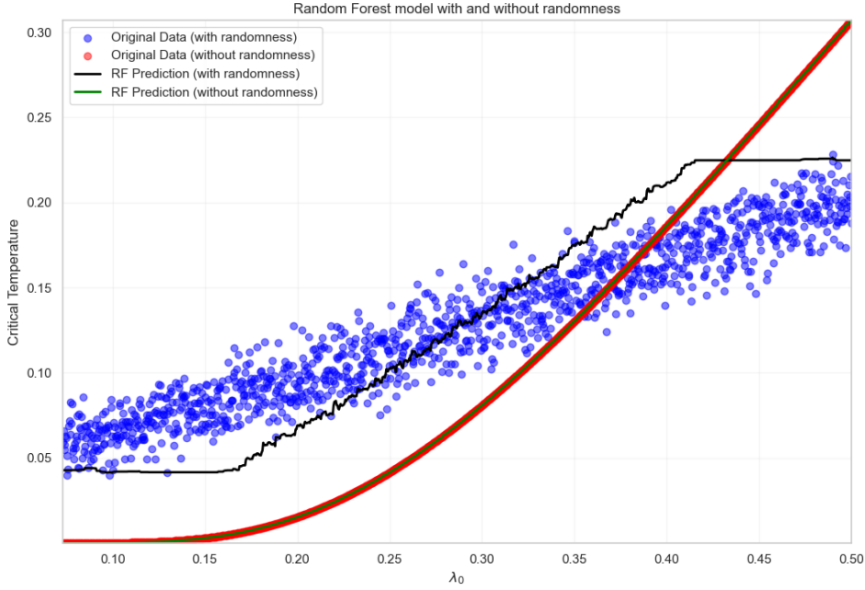


Figure 4.6: Critical temperature prediction using random forest model(with and without disorder)

In random forest analysis, we evaluated the predictive models' performance under stochastic and deterministic scenarios. Specifically, we compared the model's efficacy in predicting critical temperatures when introducing randomness versus when it is absent. The results are summarized below, providing insights into model accuracy, feature significance, and the influence of randomness. For the stochastic case, where randomness in the form of the normal distribution is incorporated through variations in the coupling constant (λ_0), the Mean Squared Error (MSE) was calculated to be 1.3807×10^{-5} , while the coefficient of determination (R^2) was 0.9929. These metrics indicate a high degree of accuracy in the model's predictions, with the R^2 value approaching 1, signifying that the model explains a substantial proportion of the variance in the critical temperatures despite the presence of randomness. In contrast, the deterministic scenario, where λ_0 remains constant, yielded an MSE of 5.2941×10^{-8} and an R^2 value of 0.9999. The significantly lower MSE and nearly perfect R^2 suggest that the model achieves near-absolute accuracy in the absence of randomness. The very low MSE underscores the minimal prediction error when the system is deterministic, reflecting an almost perfect fit of the model to the data.

Moreover, an analysis of the percentage of cases where the predicted critical temperatures in the

stochastic scenario exceed those in the deterministic scenario revealed that randomness results in higher predicted values in 84.20% of the cases. This substantial proportion highlights the pervasive impact of randomness on the system, suggesting that critical temperatures are frequently elevated when variability is present.

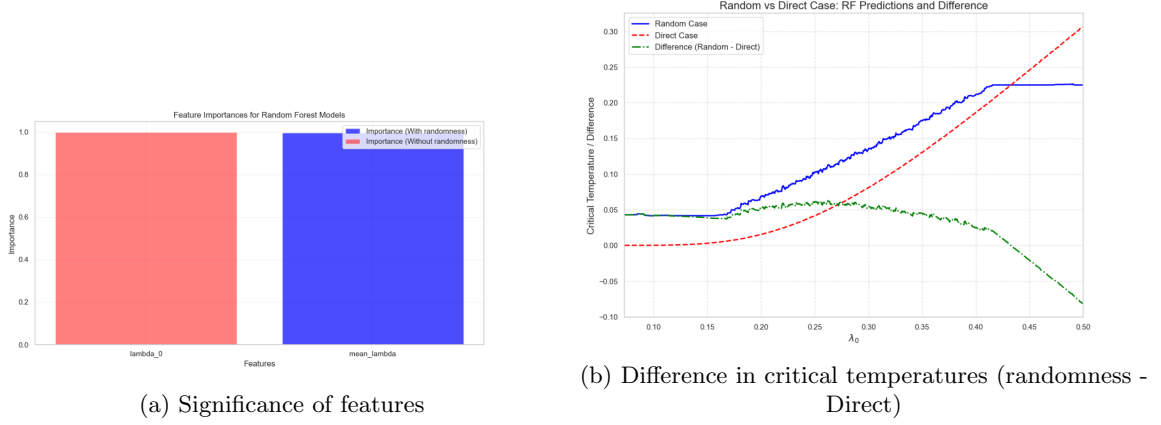


Figure 4.7: Feature importance and change in critical temperatures for random and without random variable

Random Forests provide a measure of feature importance, shown in Figure 4.7(a) allowing us to understand the relative impact of λ_0 and $mean \lambda$ on the critical temperature. Feature importance analysis further elucidates the role of various factors in the predictive models. In the stochastic case, the feature importance scores indicate that λ_0 contributes 0.0020 to the model's predictions, while $mean \lambda$ has a dominant importance score of 0.9980. This distribution suggests that, although λ_0 has a minor role in determining the critical temperatures under random conditions, the mean value of λ_0 is highly influential. Conversely, in the deterministic scenario, the feature importance scores are starkly different. λ_0 accounts for 1.0000 of the importance, indicating that it is the sole determinant of the predicted critical temperatures. The feature importance score for $mean_lambda$ is 0.0000, reflecting that the mean value does not contribute to the predictions when randomness is absent. These results underscore the pronounced effect of randomness on the predictive model's performance and the critical temperature outcomes. The stark contrast in feature importance between the two scenarios highlights the necessity of considering randomness in predictive modeling to accurately capture and understand its impact on critical temperatures. In Figure 4.7(b), the visualization offers a thorough representation of both the random and direct case predictions, as well as the disparity between the two, facilitating a more straightforward comparison and deeper understanding of how randomness influences critical temperature forecasts. Notably, the discrepancy between the predicted critical temperatures remains stable until it approaches a saturation point, beyond which a marked decline is observed. This pattern highlights the dynamic impact of randomness on the predictive model's accuracy, particularly as the system reaches critical thresholds.

4.2.2 Gaussian Processes Regression

Gaussian Processes (GPs) are a powerful class of models used extensively in regression, classification, and stochastic modeling, especially when the underlying relationships in the data are uncertain or complex. At their core, GPs offer a probabilistic approach to modeling functions, allowing for the

incorporation of uncertainty in predictions. A GP is defined as a collection of random variables, any finite number of which follow a joint Gaussian distribution. This flexibility makes GPs particularly suited for problems where the goal is to make predictions with associated confidence intervals, rather than point estimates alone. The key advantage of Gaussian Processes lies in their non-parametric nature. Unlike parametric models that assume a fixed number of parameters, GPs infer the structure of the data through a covariance function or kernel. This kernel function encodes prior assumptions about the smoothness and relationships within the data. In the context of stochastic models, where data is often generated by processes with inherent randomness or noise, GPs provide an ideal framework.

In the context of critical temperatures in superconductivity, the goal is to investigate the impact of randomness in the coupling constant (λ_0) on the critical temperature. This randomness may arise due to physical imperfections, temperature fluctuations, or other environmental factors that cause variations in λ_0 . The generated data includes cases with and without randomness in the coupling constant, making it an ideal scenario for applying Gaussian Processes. Gaussian Processes can be applied here to model the relationship between the coupling constant and critical temperatures while capturing the inherent uncertainty introduced by randomness in λ_0 . Specifically, the GP will allow for a smooth interpolation of the critical temperature over a range of λ_0 values, while simultaneously providing confidence intervals that reflect the uncertainty due to randomness. First, they enable the prediction of critical temperatures for any value of λ_0 , even for values not explicitly present in the dataset, while providing confidence intervals around these predictions. This makes GPs a robust tool for studying the variability introduced by randomness. Additionally, the flexibility of the GP allows it to adapt to complex, non-linear relationships between λ_0 and critical temperature, which may arise due to the non-linear nature of superconductivity.

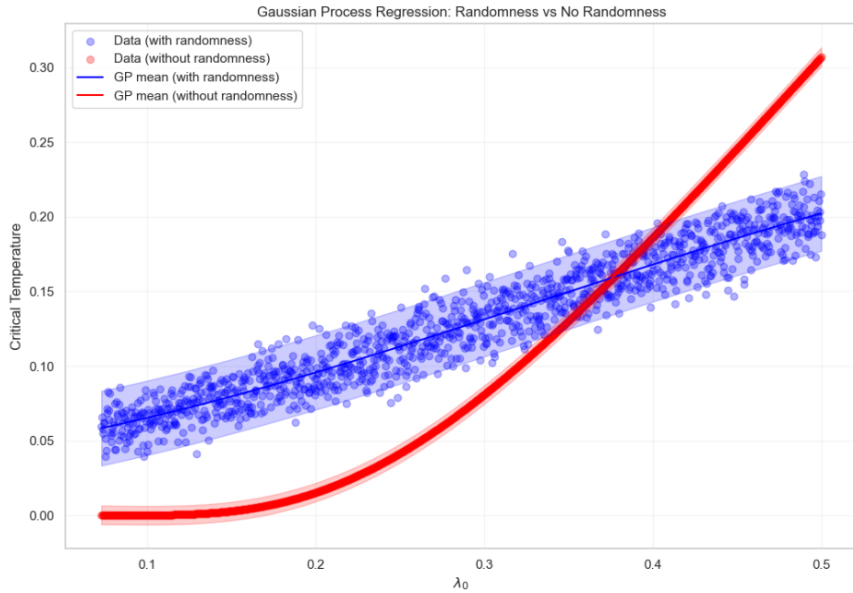


Figure 4.8: Gaussian processes regression model

The Gaussian Process Regression (GPR) model was employed to investigate the impact of ran-

domness in the coupling constant (λ_0) on the critical temperature (T_c) in superconductive systems. The model was applied to datasets representing both random and non-random variations in λ_0 , and the results offer insights into the effect of stochastic perturbations on the critical temperature. The Mean Squared Error (MSE) for the dataset with randomness was found to be 0.000158, while the R-squared value, which measures the goodness of fit, was 0.918928. These results indicate a reasonably accurate fit, though with some inherent variability due to the random fluctuations in the data. In contrast, for the dataset without randomness, the model achieved a significantly lower MSE of 0.000000, effectively demonstrating a perfect fit with an R-squared value of 0.999999. This near-perfect fit in the absence of randomness confirms that the mathematical equations governing the system are well-captured by the GPR model in deterministic conditions.

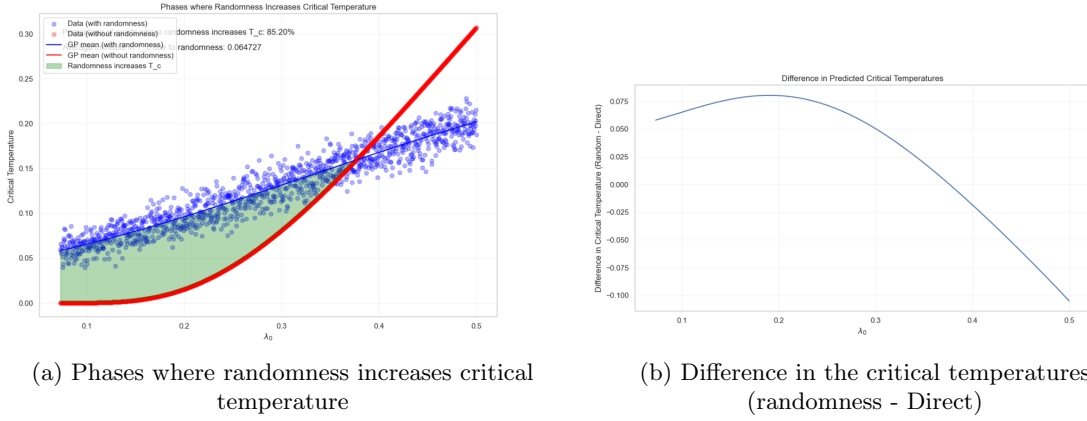


Figure 4.9: Feature importance and change in critical temperatures for random and without random variable

A key finding of this analysis is that randomness in λ_0 has a substantial effect on the critical temperature. Specifically, randomness led to an increase in T_c in 85.20% of the data points. The average increase in critical temperature due to randomness was 0.064484, while the maximum observed increase was 0.080426. These statistics suggest that the introduction of randomness in the coupling constant predominantly shifts the critical temperature upwards, likely due to stochastic fluctuations that perturb the system's behavior in a manner that enhances superconductivity in certain scenarios.

These findings provide quantitative evidence that randomness in the coupling constant has a measurable and significant impact on the critical temperature in superconductive systems. The relatively high R-squared value in the model with randomness (0.918928) suggests that the GPR model is well-suited for capturing the underlying trends despite the stochastic nature of the data. The ability of the model to effectively handle random variations in λ_0 underscores the suitability of Gaussian Processes for studying the effects of uncertainty and randomness in physical systems. The GPR model demonstrates both the robustness of superconductive systems under deterministic conditions and the pronounced effects of randomness on the critical temperature. These results contribute to a deeper understanding of how stochastic factors, such as randomness in the coupling constant, can influence the critical properties of superconductors, with potential implications for the design and control of superconductive materials in applied contexts.

4.2.3 Monte-Carlo using linear Regression

Monte Carlo methods, rooted in statistical sampling, have become indispensable tools in modeling uncertainty and randomness in various scientific fields. When applied in conjunction with linear regression models, these methods allow for robust predictions and insights, especially in cases where the underlying systems are governed by stochastic behavior. Monte Carlo simulations are particularly valuable in stochastic models, as they offer a way to quantify the variability and uncertainty in model outputs by repeatedly sampling from probability distributions of input variables. This study explores the application of Monte Carlo methods in linear regression models, particularly in the context of superconductivity, where randomness in the coupling constant (λ_0) affects the critical temperature. Monte Carlo methods are ideally suited for capturing the impact of such randomness. By repeatedly sampling from the distribution of λ_0 values, the Monte Carlo approach can generate a range of possible critical temperatures, each corresponding to a different realization of the random coupling constant. This enables the creation of a probabilistic model of T_c , offering deeper insights into how randomness in λ_0 propagates through the system and affects superconductivity. Incorporating linear regression into this framework provides an additional layer of analysis. While Monte Carlo simulations help explore the distribution of critical temperatures under random λ_0 values, a linear regression model can be used to quantify the relationship between λ_0 and T_c , with or without randomness. The regression model can also identify the linear trends or deviations that arise due to the presence of randomness in the coupling constant. In the study of superconductivity, where randomness in the coupling constant affects the critical temperature, Monte Carlo simulations provide a means to quantify the variability and uncertainty in critical temperature predictions. By integrating these simulations with linear regression, we can establish a clear relationship between λ_0 and T_c , and explore how randomness modulates this relationship. This approach not only enhances our understanding of the physical system but also offers a robust tool for making predictions in the face of uncertainty, ultimately providing more comprehensive insights into the behavior of superconductors.

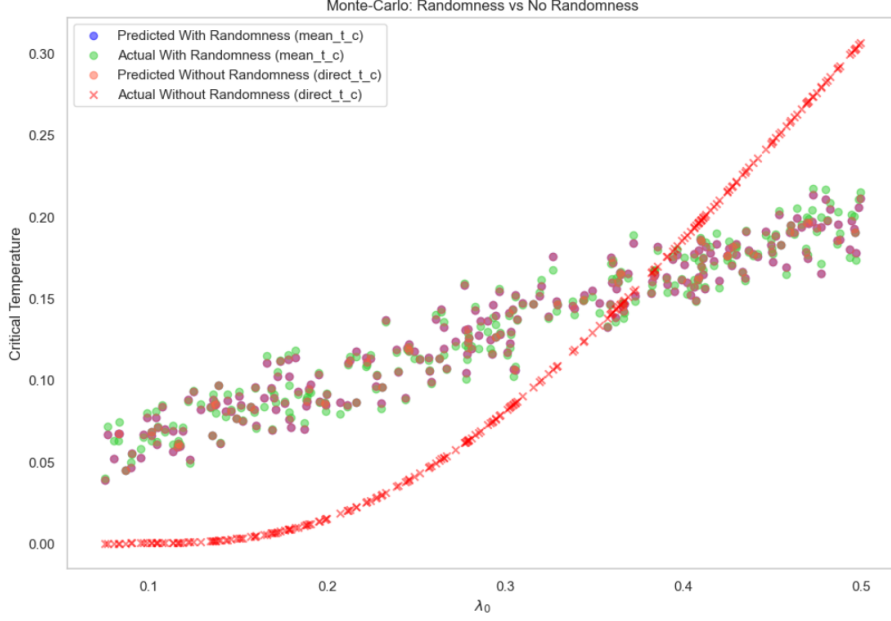


Figure 4.10: Visualization of Monte-Carlo using linear regression model

The results of the model evaluation reveal significant differences in prediction accuracy between the cases with and without randomness in the coupling constant (λ_0). Specifically, the mean squared error (MSE) for the model incorporating randomness is 1.15×10^{-5} , indicating a highly accurate fit. In contrast, the model without randomness exhibits a considerably higher MSE of 4.27×10^{-3} , suggesting a less precise prediction of critical temperature.

These findings underscore the importance of accounting for randomness in the coupling constant when modeling the critical temperature in superconductors. The much lower MSE in the model with randomness implies that the inclusion of stochastic variations in λ_0 leads to a more accurate and reliable representation of the system's behavior, capturing nuances that the deterministic model without randomness fails to reflect.

Our results indicate that approximately 70.75% of the λ_0 values analyzed showed an increase in T_c due to the introduction of randomness. Quantitatively, the average increase in T_c attributed to randomness was 0.0280, while the maximum increase reached 0.1131. Notably, this maximum increase occurred at a λ_0 value of 0.1980, highlighting the significant effect of randomness at this coupling constant level. Integrating Monte Carlo simulations with linear regression enabled us to effectively quantify the variability and uncertainty in T_c predictions. This comprehensive approach not only elucidated the relationship between λ_0 and T_c but also identified critical points where randomness exerted the most influence. Our findings emphasize the necessity of accounting for randomness in the coupling constant when analyzing superconducting materials, enhancing our understanding of the underlying physical processes and improving predictive accuracy under uncertain conditions.

4.2.4 Variational Autoencoder

Variational autoencoders (VAEs) represent a powerful class of generative models that utilize probabilistic inference to capture complex data distributions. By mapping high-dimensional data into a lower-dimensional latent space, VAEs facilitate the learning of underlying representations while

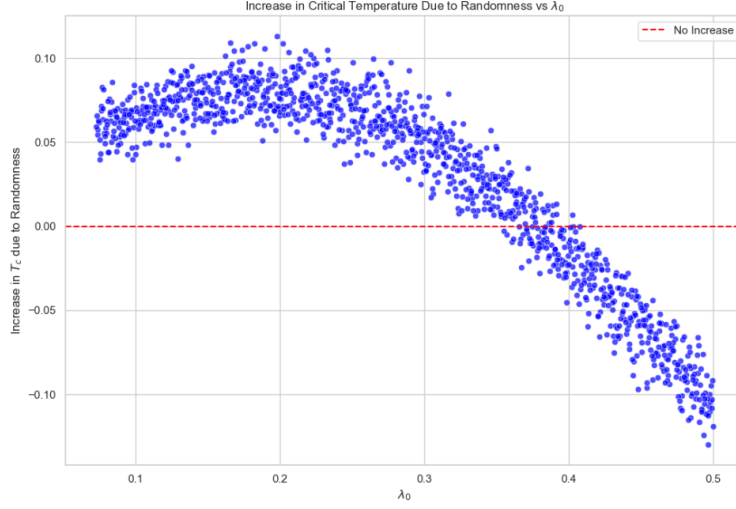


Figure 4.11: Change in critical temperatures due to randomness

allowing for the generation of new data samples. This framework is particularly useful in stochastic modeling, where the inherent variability in data can be modeled through a probabilistic approach. In the context of systems influenced by randomness, such as critical temperatures in superconductivity, VAEs can be employed to discern the subtle effects of stochastic perturbations, particularly those arising from variations in coupling constants like λ_0 . VAEs can generate synthetic datasets that simulate different scenarios of randomness, allowing for a comprehensive analysis of how fluctuations in the coupling constant affect the critical temperatures. This generative capability provides a robust framework for hypothesis testing and scenario analysis, facilitating a deeper understanding of the stochastic processes at play. Ultimately, by leveraging variational autoencoders, our research can contribute to a nuanced comprehension of the intricate dynamics of superconductivity, illuminating the role of randomness in determining critical phenomena.

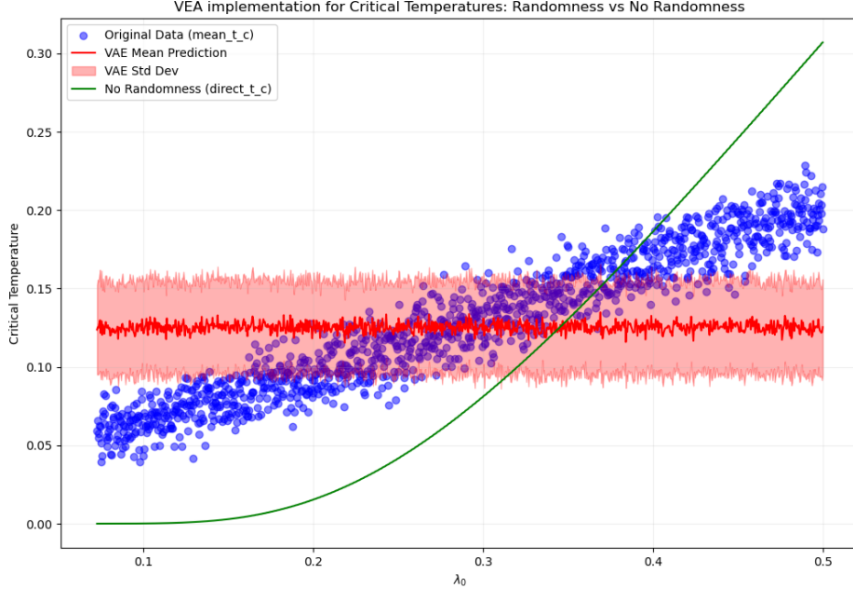


Figure 4.12: Critical temperature for random and without randomness using VEA model

1,000 samples were used to compare the critical temperatures predicted by the Variational Autoencoder (VAE) model with those derived directly from the deterministic case, without the influence of randomness. Notably, in 63.50% of the cases, the mean critical temperature predicted by the VAE exceeded the corresponding critical temperature obtained in the deterministic scenario (T_c^{direct}). This observation suggests that the VAE, by modeling the inherent variability introduced by randomness, often predicts slightly higher critical temperatures than those calculated without random perturbations. The Mean Squared Error (MSE) calculated between the VAE mean predictions and the direct critical temperatures is 0.010007. The VAE model introduces only minor deviations from the deterministic case, supporting the conclusion that randomness in the coupling constant has a small but measurable impact on critical temperatures in the superconductivity.

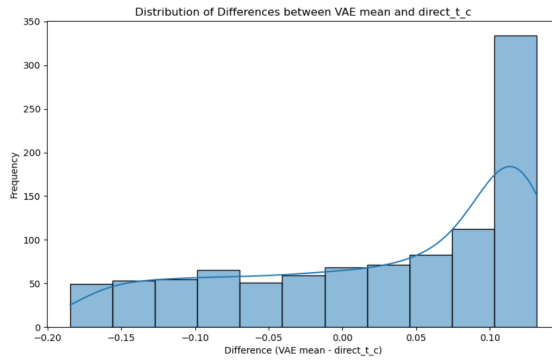


Figure 4.13: Frequency of Distribution of differences of predicted and direct critical temperature

Furthermore, the average difference between the VAE mean predictions and T_c^{direct} in cases where the VAE mean is greater was found to be 0.090885. This indicates that, on average, the VAE predicts a modest but non-negligible increase in critical temperatures when accounting for stochastic variations in the coupling constant (λ_0). These results underscore the utility of the VAE in capturing and reflecting the effects of randomness in the system, providing a more comprehensive understanding of the relationship between stochastic perturbations and critical phenomena in superconductivity.

Chapter 5

Conclusion and Future Direction

In this study, we analyzed a dataset generated from a mathematically derived stochastic equation with high precision, designed to simulate the behavior of superconductors under varying levels of disorder. The coupling constant (λ_0) was chosen within a range that corresponds to the typical behavior of superconductors, while disorder was introduced through a normal distribution to reflect the real-world presence of impurities in superconducting materials.

Our analysis revealed several important observations. When significant disorder is present in superconductors with a small coupling constant in the range of $0 \leq \lambda_0 \leq 0.3657$, the critical temperature tends to increase by approximately 70.75%. This increase suggests that the actual value of the coupling constant in disordered systems deviates from the behavior predicted for ideal (disorder-free) conditions, leading to enhanced critical temperatures.

To validate these findings, we employed multiple machine learning models, each of which corroborated the general trend of increased critical temperatures in disordered systems. The Random Forest model indicates approximately 84.20% of the time, the critical temperature in a disordered state is higher than that in the ideal, disorder-free state. Gaussian Process regression further supports this trend, suggesting that in 85.20% of cases, disordered superconductors exhibit higher critical temperatures within the specified range of coupling constants. Monte Carlo simulations with regression analysis imply that the coupling constant positively influences the critical temperature approximately 70.75% of the time until a saturation point is reached, beyond which the critical temperature begins to decrease relative to the ideal state. This non-linear behavior highlights the complex interaction between disorder and the coupling constant. Furthermore, we employed a Variational Autoencoder (VAE) model to capture the intricate behavior and uncertainty within the data. The VAE model predicts that in 63.50% of cases, disordered superconductors exhibit higher critical temperatures than their disorder-free counterparts.

Overall, these observations, supported by both statistical analysis and machine learning models, provide strong evidence for the positive influence of disorder on superconductivity, particularly when the coupling constant is approximately less than 0.35. Our results suggest that introducing a controlled amount of disorder can significantly enhance the critical temperature of superconductors, a finding that could have important implications for the development of high-performance superconducting materials. Given that the dataset used in this study was generated from mathematical equations with added disorder, experimental validation is a crucial next step. Future research should focus on testing the predictions of increased critical temperatures in real-world superconductors by introducing controlled levels of disorder. Collaborations with experimentalists to synthesize materials with varying impurity concentrations and to measure their corresponding critical temperatures

would provide essential validation of our findings. In this study, the disorder was modeled using a normal distribution to represent impurities in the superconductors. Future investigations could explore more complex and realistic models of disorder, such as Anderson localization or non-Gaussian distributions, which may more accurately reflect the impurities and defects observed in real materials. A deeper understanding of how different types of disorders affect superconductivity could lead to new insights into the underlying mechanisms governing the behavior of disordered superconductors. While our mathematical model was tailored to certain types of superconductors, it is important to recognize that different materials may exhibit unique responses to disorder. Future work could involve generating new datasets based on models that simulate various classes of superconductors, including high-temperature and unconventional superconductors. Machine learning models could then be retrained on these expanded datasets to determine whether the trends observed in this study are consistent across different materials. Although a range of machine learning models were utilized in this work, future research could explore additional advanced techniques, such as deep learning, reinforcement learning, or hybrid approaches that integrate physics-based models with machine learning. These techniques could potentially capture even more intricate dependencies between disorder and critical temperature. Furthermore, methods such as active learning or unsupervised learning may uncover new patterns and hidden insights within the data, enhancing our understanding of the role of disorder in superconductivity. Lastly, future studies could benefit from integrating the current mathematical models with more detailed quantum mechanical simulations, such as density functional theory (DFT) or quantum Monte Carlo methods. Incorporating these quantum mechanical approaches would allow for a more accurate representation of the electron-disorder interactions, thereby improving the robustness and reliability of our predictions.

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Appendix A

Data Generation

This section outlines the code used to calculate the critical temperature (t_c) corresponding to varying values of the thermal conductivity parameter (λ_0). The methodology employs numerical root-finding techniques to achieve this.

```
def find_t_c(lambda_0):
    try:
        result = optimize.root_scalar(equation, args=(lambda_0,), bracket=[1e-6, 1e6], method='brentq')
        return result.root
    except ValueError:
        return np.nan

np.random.seed(42) # for reproducibility
lambda_0_values = np.linspace(0.068, 0.5, 1200)
np.random.shuffle(lambda_0_values) # Shuffle to randomize the order

sigma = np.sqrt(0.1)
n_samples = 1

dataset = []

for lambda_0 in lambda_0_values:
    # Generate random lambda value
    lambda_value = np.random.normal(lambda_0, sigma)
    lambda_value = np.clip(lambda_value, 0.068, 0.5)

    # Calculate corresponding t_c value
    t_c_value = find_t_c(lambda_value)

    # Store data
    dataset.append({'lambda_0': lambda_0, 'lambda': lambda_value, 't_c': t_c_value})
```

Figure A.1: Implementation for generating dataset

The function `find_t_c` aims to determine the critical temperature t_c for a given value of λ_0 . This is accomplished using the `optimize.root_scalar` function from the SciPy library, which employs Brent's method (specified by `method='brentq'`).

The function accepts a single argument, λ_0 , and attempts to find the root of the equation, which is presumed to be predefined elsewhere in the code. To create a dataset for analysis, we generate 1200 unique values for λ_0 using the `numpy.linspace` function. The range is set from 0.068 to 0.5, with a deliberate lower limit of 0.068 to ensure that valid thermal conductivity values are utilized, as values below this threshold may lead to non-physical results. The generated values are then shuffled to introduce randomness in the dataset, enhancing the robustness of subsequent analyses.

Appendix B

Machine learning Models

B.1 Random forest Model

```
# Split the data
X_train, X_test, y_random_train, y_random_test, y_direct_train, y_direct_test = train_test_split(
    X, y_random, y_direct, test_size=0.2, random_state=42)

# Train Random Forest models
rf_random = RandomForestRegressor(n_estimators=100, random_state=42)
rf_direct = RandomForestRegressor(n_estimators=100, random_state=42)

rf_random.fit(X_train, y_random_train)
rf_direct.fit(X_train, y_direct_train)

# Make predictions
y_random_pred = rf_random.predict(X_test)
y_direct_pred = rf_direct.predict(X_test)

# Calculate metrics
mse_random = mean_squared_error(y_random_test, y_random_pred)
r2_random = r2_score(y_random_test, y_random_pred)
mse_direct = mean_squared_error(y_direct_test, y_direct_pred)
r2_direct = r2_score(y_direct_test, y_direct_pred)
```

Figure B.1: Random Forest implementation

The predictor variables, X , consist of two features: λ_0 and λ_{mean} , which serve as input variables to the Random Forest models. Two different target variables were used: y_{random} (the mean critical temperature containing randomness $mean_{t_c}$) and y_{direct} (critical temperature with no randomness $direct_{t_c}$).

The data was split into training and testing sets using an 80/20 split ratio, with the `train_test_split` function from the `sklearn` library. This function splits the input features and target variables for the two models into corresponding training and testing subsets. A random seed (`random_state=42`) was set to ensure the reproducibility of the results.

B.1.1 Model Training

Two Random Forest models were employed: one for predicting y_{random} and another for predicting y_{direct} . Both models were initialized using the `RandomForestRegressor` class from `sklearn`, with the number of estimators (`n_estimators`) set to 100, which specifies that each model will construct

100 decision trees as part of the ensemble. A fixed random state (`random_state=42`) was provided to ensure deterministic model training.

Each model was trained on the training subset of the data using the `fit` method, which fits the Random Forest model to the respective target variables.

B.1.2 Model Prediction

After training, predictions were made on the test data using the `predict` method. Both models predicted the respective target variables on the testing set X_{test} . The predicted values, $y_{\text{random_pred}}$ for y_{random} and $y_{\text{direct_pred}}$ for y_{direct} , represent the Random Forest’s estimation of the target variables based on the test input features.

$$y_{\text{random_pred}} = \text{RandomForest}_{\text{random}}.\text{predict}(X_{\text{test}})$$

$$y_{\text{direct_pred}} = \text{RandomForest}_{\text{direct}}.\text{predict}(X_{\text{test}})$$

This approach leverages the ensemble nature of Random Forest models to capture complex relationships between the input features (λ_0 and λ_{mean}) and the target thermal conductivity variables. Random Forest is well-suited for regression tasks due to its ability to handle non-linearities and interactions between features, while also providing robust predictions by averaging over multiple decision trees.

B.2 Gaussian Processes Regression

```
# Split the data into training and testing sets
X_train, X_test, y_random_train, y_random_test, y_direct_train, y_direct_test = train_test_split(
    X, y_random, y_direct, test_size=0.2, random_state=42)

# Define the kernel for the Gaussian Process
kernel = RBF() + WhiteKernel()

# Create and fit the Gaussian Process models
gp_random = GaussianProcessRegressor(kernel=kernel, n_restarts_optimizer=10, random_state=42)
gp_direct = GaussianProcessRegressor(kernel=kernel, n_restarts_optimizer=10, random_state=42)

gp_random.fit(X_train, y_random_train)
gp_direct.fit(X_train, y_direct_train)

# Make predictions
X_pred = np.logspace(np.log10(X.min()), np.log10(X.max()), 1000).reshape(-1, 1)
y_pred_random, sigma_random = gp_random.predict(X_pred, return_std=True)
y_pred_direct, sigma_direct = gp_direct.predict(X_pred, return_std=True)
```

Figure B.2: Implementation of Gaussian processes model

The Gaussian Process models were constructed using a combination of two kernels:

- The Radial Basis Function (RBF) kernel, commonly used to model smooth functions.
- The WhiteKernel, which accounts for noise in the observations.

The choice of this composite kernel ensures that the model captures both the underlying smooth trends and any noise or variability in the data.

$$\text{Kernel} = \text{RBF}() + \text{WhiteKernel}()$$

B.2.1 Model Training

Two Gaussian Process regression models were created, one for each target variable (y_{random} and y_{direct}). The models were implemented using the `GaussianProcessRegressor` class from `sklearn`. The `n_restarts_optimizer` parameter was set to 10, allowing the optimizer to restart the fitting process multiple times to avoid local minima and ensure a more reliable estimate of the model parameters.

The models were then trained using the training datasets for both target variables:

$$\begin{aligned} & \text{GPR}_{\text{random}}.\text{fit}(X_{\text{train}}, y_{\text{random_train}}) \\ & \text{GPR}_{\text{direct}}.\text{fit}(X_{\text{train}}, y_{\text{direct_train}}) \end{aligned}$$

B.2.2 Uncertainty Estimation

Following the training phase, predictions were made over a grid of input values, X_{pred} , which were generated by logarithmically spacing 1000 points between the minimum and maximum values of the feature space X . This log-scale transformation ensures better coverage of the feature space, particularly when there is a wide dynamic range in the data.

$$X_{\text{pred}} = \text{np.logspace}(\log_{10}(X_{\text{min}}), \log_{10}(X_{\text{max}}), 1000).\text{reshape}(-1, 1)$$

The trained Gaussian Process models were then used to predict the target variables over this grid. Additionally, the models provide an estimate of the uncertainty (σ) associated with each prediction, which represents the model's confidence in its predictions.

$$\begin{aligned} y_{\text{pred_random}}, \sigma_{\text{random}} &= \text{GPR}_{\text{random}}.\text{predict}(X_{\text{pred}}, \text{return_std} = \text{True}) \\ y_{\text{pred_direct}}, \sigma_{\text{direct}} &= \text{GPR}_{\text{direct}}.\text{predict}(X_{\text{pred}}, \text{return_std} = \text{True}) \end{aligned}$$

The standard deviations σ_{random} and σ_{direct} quantify the model's predictive uncertainty for the mean and direct thermal conductivity predictions, respectively. These uncertainties are crucial for understanding the reliability of the predictions made by the Gaussian Process models.

B.3 Monte-Carlo with linear regression

```
X_train, X_test, Y_mean_train, Y_mean_test = train_test_split(X, Y_mean_tc, test_size=0.2, random_state=42)

# Train a Linear Regression model
lr_model = LinearRegression()
lr_model.fit(X_train, Y_mean_train)

# Predictions
predictions_mean = lr_model.predict(X_test)
model_mean_mse = mean_squared_error(Y_mean_test, predictions_mean)
```

Figure B.3: implementation of Monte-Carlo with regression

The model was trained, and predictions were made on the generated data X_{test} . The predicted values $\hat{Y}_{\text{mean}}$ were then compared to the true test values $Y_{\text{mean_test}}$ to assess the performance of the linear regression model:

$$\hat{Y}_{\text{mean}} = \text{lr_model.predict}(X_{\text{test}})$$

The model’s performance was evaluated using the Mean Squared Error (MSE), a widely-used metric that quantifies the average squared difference between the predicted values and the true values. The MSE provides insights into how well the linear model captures the underlying relationships in the data, given the presence of randomness in the target variable:

$$\text{MSE}_{\text{mean}} = \text{mean_squared_error}(Y_{\text{mean_test}}, \hat{Y}_{\text{mean}})$$

Finally, the results were printed to report the error in predictions, reflecting the effect of randomness in the target variable:

```
print("With Randomness MSE: ", MSE_mean)
```

This process serves as the foundation for the Monte Carlo simulation, in which randomness is introduced into the target variable t_c across multiple simulations to assess its effect on model accuracy. The Linear Regression model is utilized as the core predictor in these simulations, forecasting critical temperatures under the influence of stochastic factors. By evaluating the Mean Squared Error across different random splits and feature sets, we gain a deeper understanding of the variability and uncertainty associated with the predictions.

B.4 Variational Autoencoder

The encoder is defined as a sequential neural network with two hidden layers consisting of 64 and 32 units, respectively. Each hidden layer is followed by a Rectified Linear Unit (ReLU) activation function. The output of the encoder is split into two parts: the mean (μ) and the logarithmic variance ($\log \sigma^2$) of the latent variable distribution. These two outputs are obtained by partitioning the final output of the encoder into two equal-sized tensors. The size of the latent space is determined by the hyperparameter `latent_dim`.

```
# Scale the data
scaler_X = StandardScaler()
scaler_y = StandardScaler()
X_scaled = scaler_X.fit_transform(X)
y_scaled = scaler_y.fit_transform(y)

# Split the data
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y_scaled, test_size=0.2, random_state=42)

# Convert to float32
X_train = X_train.astype(np.float32)
y_train = y_train.astype(np.float32)
X_test = X_test.astype(np.float32)
y_test = y_test.astype(np.float32)

# Define the VAE model
latent_dim = 2
```

Figure B.4: Scaling and defining datatypes

B.4.1 Encoder

The encoder can be mathematically represented as follows:

$$h = \text{ReLU}(W_1x + b_1)$$

where h is the output of the first hidden layer, W_1 is the weight matrix, x is the input vector, and b_1 is the bias vector. The latent representation is then defined as:

$$z_{\text{latent}} = [\mu, \log \sigma^2] = \text{Dense}(h)$$

```
# Define the VAE model
latent_dim = 2

class VAE(keras.Model):
    def __init__(self, latent_dim):
        super(VAE, self).__init__()
        self.latent_dim = latent_dim
        self.encoder = tf.keras.Sequential([
            layers.Dense(64, activation="relu"),
            layers.Dense(32, activation="relu"),
            layers.Dense(latent_dim + latent_dim),
        ])
        self.decoder = tf.keras.Sequential([
            layers.Dense(32, activation="relu"),
            layers.Dense(64, activation="relu"),
            layers.Dense(1),
        ])

    def encode(self, x):
        mean, logvar = tf.split(self.encoder(x), num_or_size_splits=2, axis=1)
        return mean, logvar

    def reparameterize(self, mean, logvar):
        eps = tf.random.normal(shape=mean.shape)
        return eps * tf.exp(logvar * .5) + mean

    def decode(self, z):
        return self.decoder(z)

    def call(self, inputs):
        mean, logvar = self.encode(inputs)
        z = self.reparameterize(mean, logvar)
        return self.decode(z), mean, logvar
```

Figure B.5: VAE model implementation

where z_{latent} consists of the mean μ and the logarithmic variance $\log \sigma^2$, produced by a dense layer that maps h to the latent space.

B.4.2 Decoder

The output of the decoder reconstructs the input data, which can be expressed as:

$$h' = \text{ReLU}(W_2 z + b_2)$$

Here, h' is the output of the first hidden layer in the decoder, W_2 is the weight matrix, z is the latent variable representation, and b_2 is the bias vector. Finally, the reconstructed input is given by:

$$\hat{x} = \text{Dense}(h')$$

where $\hat{x}$ denotes the reconstructed output produced by the dense layer of the decoder.

This architecture enables the model to learn a probabilistic representation of the input data while facilitating effective reconstruction during the decoding phase.

B.4.3 Reparameterization

The Variational Autoencoder (VAE) employs reparameterization to facilitate backpropagation through the stochastic latent variable. Specifically, the latent variable z is computed as follows:

$$z = \mu + \epsilon \cdot \exp\left(\frac{1}{2} \log \sigma^2\right)$$

where $\epsilon \sim \mathcal{N}(0, I)$ is a random sample drawn from a standard normal distribution, and $\log \sigma^2$ represents the log variance output from the encoder.

This approach allows the latent variables to be sampled while maintaining a differentiable operation, which is crucial for optimizing the model through gradient-based methods.

B.4.4 VAE execution

The `call` function defines the forward pass of the model. During training, the input data x is first encoded to obtain the latent mean μ and log variance $\log \sigma^2$. These parameters are subsequently utilized to sample a latent vector z using the reparameterization trick. Finally, the decoder reconstructs the input from the latent vector z .